

Data Sources Documentation

This project utilizes publicly available statistical data to build an intelligent energy consumption assistant. Rather than tracking individual devices, the system compares the user's monthly electricity bill and contextual inputs (e.g., housing type, ownership, and regional temperature) against official statistical baselines. By integrating this data, the LLM can generate highly personalized awareness messages. The table below outlines all data sources evaluated, indicating which were adopted and which were excluded.

Data Source Name	Link	Type of Data Provided	Availability (Public / Restricted)	Challenges in Accessing it	Contacted the Provider?	Response Received
GASTAT Household Energy Statistics (2024)	Link	<i>(Used)</i> Aggregated average operating hours for major electrical uses (AC, heating, water heating, cooking) based on housing type, ownership, and region.	Public	None. Directly downloadable in tabular format.	No	N/A
Open-Meteo Historical Weather API	Link	<i>(Used)</i> Historical monthly temperature data to determine seasons and map them to consumption patterns.	Public	Requires processing and mapping to align with GASTAT's seasonal definitions.	No	N/A
National Electricity Provider (SEC)	N/A	<i>(No response)</i> Detailed, non-identifiable residential electricity consumption data for academic research.	Restricted	Highly confidential. Requires multiple departmental and college approvals; access is not guaranteed and takes time.	Yes	No response yet. Waiting for college approval letters.

KAPSARC	N/A	(No response) Datasets regarding Saudi residential electricity consumption.	Restricted	Data is usually restricted to official research partners and requires formal requests.	Yes	No response yet.
Open Data Portal (Gov.sa)	Link	(Not Useful) Total annual electricity consumption in the Central Region.	Public	Lacks household-level details, appliance indicators, or actionable attributes needed for the app.	No	N/A
SAMA Economic Reports	Link	(Not Useful) Regional electricity consumption financial reports.	Public	Highly aggregated data. Cannot be used to extract household patterns.	No	N/A
Kaggle: Household Electricity Usage & Appliance-wise Hourly Consumption	Link 1 Link 2	(Not Useful) Hourly electricity consumption data recorded at the individual appliance level for multiple households (10,000+ rows).	Public	Excluded: The dataset provides hourly, device-level tracking, which does not align with the application's approach of analyzing overall monthly bills.	No	N/A
Kaggle: Household Energy Consumption	Link	(Not Useful) Household energy records (90,000 rows) including features like Temp, Has_AC, and Peak Hours.	Public	Excluded: Severely limited timeframe. The dataset only contains data for one month (April), which eliminates the ability to analyze seasonal consumption trends (Winter vs. Summer) required for the app.	No	N/A